DIABETES DIETARY GUIDELINES

Clinical Nutrition Reference for Meal Planning & RAG-Based Diet Generation

1. Overview of Diabetes

Type 1 Diabetes: An autoimmune condition in which the pancreatic beta cells are destroyed, resulting in absolute insulin deficiency. Requires exogenous insulin. Onset is typically in childhood or early adulthood.

Type 2 Diabetes: A metabolic disorder characterized by progressive insulin resistance and relative insulin deficiency. Strongly associated with obesity, sedentary lifestyle, and genetic predisposition. Accounts for ~90–95% of all diabetes cases worldwide.

Insulin and Glucose: Insulin is a hormone secreted by pancreatic beta cells that facilitates cellular glucose uptake. In its absence or with reduced sensitivity, blood glucose rises (hyperglycemia), causing systemic damage over time to kidneys, eyes, nerves, and cardiovascular system.

2. Core Dietary Goals

- **Stabilize Blood Sugar:** Avoid rapid glucose fluctuations by choosing complex carbohydrates and fiber-rich foods.
- **Avoid Postprandial Spikes:** Target postprandial glucose <140 mg/dL at 2 hours. Limit high-GI foods and refined sugars.
- **Improve Insulin Sensitivity:** Reduce visceral adiposity through calorie moderation, regular physical activity, and anti-inflammatory dietary patterns.
- **Achieve Target HbA1c:** Dietary management aims to achieve HbA1c <7% (ADA guideline). Every 1% reduction decreases microvascular complication risk by ~25%.

3. Carbohydrate Management

Carbohydrates have the greatest direct impact on blood glucose among all macronutrients. Recommended carbohydrate intake: **45–55% of total daily energy** (ADA/ICMR). Emphasis must be placed on carbohydrate *quality*, not only quantity.

Carb Type	Examples	Effect on Blood Sugar	Recommendation
Simple/Refined	White bread, sugar, maida, candy, soft drinks	Rapid spike (high GI)	AVOID
Complex (Whole)	Oats, brown rice, quinoa, whole wheat	Slow, gradual rise	PREFER
High-Fiber	Lentils, beans, vegetables, oats	Blunts glucose spike	INCLUDE DAILY
Resistant Starch	Cooled rice, green banana, raw oats	Minimal glucose impact	INCLUDE

FIBER RULE: Target 25–35 g/day total dietary fiber (ADA). Soluble fiber (oats, psyllium, legumes) slows gastric emptying and reduces postprandial hyperglycemia. Insoluble fiber (wheat bran, vegetables) improves gut motility.

4. Glycemic Index (GI) — CRITICAL REFERENCE

The Glycemic Index (GI) ranks foods (0–100) based on how quickly they raise blood glucose compared to pure glucose (GI=100). **Low GI (<55)** foods are preferred for diabetics. **Glycemic Load (GL)** = (GI × carb grams per serving) / 100; GL <10 is considered low.

Food Item	GI Value	Category	Recommendation
Glucose (reference)	100	High	Avoid
White bread (maida)	75	High	Avoid
White rice (cooked)	72	High	Limit / portion control
Watermelon	72	High	Limit
Ripe banana	62	Medium	Limit portions
Whole wheat bread	54	Low-Medium	Moderate use
Brown rice	50	Low-Medium	Prefer over white rice
Sweet potato (boiled)	46	Low	Include
Oats (rolled)	55	Low-Medium	Include (breakfast)
Apple	38	Low	Include
Chickpeas (chana)	28	Low	Include
Lentils (moong/masoor)	25–32	Low	Include daily
Kidney beans (rajma)	24	Low	Include
Barley	28	Low	Include
Milk (full-fat)	31	Low	Include (moderate)
Plain yogurt	14–17	Low	Include daily
Vegetables (non-starchy)	<15	Very Low	Unrestricted

5. Protein Guidelines

Recommended intake: 1.0–1.5 g/kg body weight/day for most T2DM patients (ICMR 2020). Higher protein intakes (1.2–1.8 g/kg) may be appropriate in obese individuals for weight loss.

Protein causes minimal acute glycemic response. It stimulates insulin secretion modestly and promotes satiety, reducing overall carbohydrate intake. Protein helps preserve lean muscle mass during calorie-restricted weight loss programs.

Note: *In patients with diabetic nephropathy (kidney disease), restrict protein to 0.6–0.8 g/kg/day. Always assess renal function (eGFR, urine albumin) before recommending protein targets.*

6. Fat Guidelines

Fat Type	Sources	Effect	Target
Monounsaturated (MUFA)	Olive oil, avocado, almonds, groundnuts	Improves insulin sensitivity	Prefer — 15–20% total energy
Polyunsaturated (PUFA)	Fish (omega-3), walnuts, flaxseed, sunflower oil	Reduces inflammation, improves lipids	Include — 10–15%
Saturated Fat	Butter, ghee, coconut oil, red meat, full-fat dairy	Worsens insulin resistance	Limit — <7% total energy
Trans Fat	Vanaspati, margarine, fried snacks, bakery items	Increases CVD risk, worsens IR	AVOID completely

7. Meal Timing & Frequency

- Eat **3 main meals + 1–2 small snacks** per day; avoid skipping meals.
- Space meals **3–4 hours apart** to prevent hypoglycemia and excessive hunger.
- **Avoid prolonged fasting (>6 hours)**; can trigger reactive hypoglycemia or compensatory overeating.
- Eat **breakfast within 1–2 hours of waking** to stabilize morning glucose.
- Keep **carbohydrate portions consistent** across meals for better glycemic control.
- Limit caloric intake after 8 PM; evening meals should be lighter.

8. Food Lists — Allowed vs. Avoid

■ **ALLOWED / RECOMMENDED**

- Oats, barley, broken wheat (dalia)
- Brown rice, red rice, millets (ragi, jowar)
- Lentils: moong, masoor, chana, rajma, urad
- All non-starchy vegetables (spinach, broccoli, methi, gourd)
- Eggs, grilled/poached chicken, fish (salmon, mackerel)
- Low-fat yogurt, buttermilk, paneer (moderate)
- Nuts: almonds (5–6), walnuts (2–3) per day
- Seeds: flaxseed, chia, fenugreek (methi) seeds
- Fresh fruits: apple, pear, guava, papaya (small portions)
- Green tea, plain water, vegetable soups

■ **AVOID / RESTRICT**

- Sugar, jaggery, honey (except medical use)
- White rice (unrestricted), maida (refined flour)
- Bread, biscuits, cakes, pastries, namkeen
- Sugary beverages: cola, juices, energy drinks
- Deep-fried foods: samosa, vada, puri, bhatura
- Full-fat ice cream, sweetened yogurt, kheer
- Processed meats: sausages, salami
- Alcohol (esp. beer; if consumed: max 1 unit/day with food)
- High-GI fruits: mango, chiku, grapes (large qty)
- Packaged cereals with added sugar

9. Sample Diet Plans (1800 kcal / Day)

Sample Plan A — Vegetarian (South Indian)

Meal	Time	Food Items	Approx. Kcal
Early Morning	6:30 AM	Methi water (1 glass) + 5 soaked almonds	60
Breakfast	8:00 AM	Oats upma (1 cup) + 1 boiled egg or moong sprouts (50g)	320
Mid-Morning	10:30 AM	1 small guava or apple + buttermilk (200 ml)	120
Lunch	1:00 PM	Brown rice (1 cup) + sambar (1 cup) + sabzi + salad + curd	520
Evening Snack	4:30 PM	Roasted chana (30g) + green tea	130
Dinner	7:30 PM	2 jowar/ragi roti + dal (1 cup) + vegetable sabzi	450
Bedtime	9:30 PM	Warm turmeric milk (low-fat, 150 ml)	80
TOTAL		~1680–1800 kcal	~50g fiber, ~75g protein

Sample Plan B — Vegetarian (North Indian)

Meal	Time	Food Items	Approx. Kcal
Early Morning	6:30 AM	Warm water + 1 tsp psyllium (isabgol)	10
Breakfast	8:00 AM	2 whole wheat rotis + vegetable poha (1 cup) + curd	380
Mid-Morning	11:00 AM	1 small pear + handful of walnuts (2–3 halves)	110
Lunch	1:30 PM	2 bajra/wheat roti + rajma (1 cup) + salad + buttermilk	540
Evening	5:00 PM	Sprout chaat (50g) + lemon juice, no sugar	140
Dinner	7:30 PM	Dalia khichdi (1 cup) + sauteed spinach + dal soup	430
TOTAL		~1610–1800 kcal	~45g fiber, ~72g protein

10. Clinical Notes & Monitoring

- **Fiber:** Target 25–35 g/day. Introduce gradually to avoid GI discomfort. Psyllium husk, oats, and legumes are excellent soluble fiber sources.
- **Hydration:** 8–10 glasses of water/day (2–2.5 L). Avoid sugary drinks. Infused water (jeera, cucumber) is acceptable.
- **Blood Glucose Monitoring:** Fasting target 80–130 mg/dL; 2-hr postprandial <180 mg/dL (ADA). Self-monitoring guides food choices.
- **HbA1c Monitoring:** Every 3 months until controlled; every 6 months once stable. Target <7%.
- **Sodium:** Limit to <2300 mg/day, especially in hypertensive diabetics.
- **Alcohol:** If consumed, max 1 unit/day with food; monitor for hypoglycemia.
- **Supplement consideration:** Vitamin D, B12, and magnesium deficiencies are common in T2DM patients and may worsen insulin resistance — assess and supplement if needed.

References: ADA Standards of Medical Care in Diabetes 2024 • ICMR Dietary Guidelines for Indians 2020 • WHO/FAO GI Tables • Diabetes Care Journal